Phishing Email Detection & Awareness System

1. Introduction

Phishing is one of the most common and dangerous cyber threats faced by individuals and organizations today. It is a type of social engineering attack where attackers impersonate legitimate entities to trick users into revealing sensitive information such as passwords, banking details, or OTPs.

With the rapid growth of digital communication, email has become a primary target for phishing attacks. These attacks can lead to financial loss, data breaches, and reputational damage. Therefore, detecting phishing emails and creating awareness among users is critical.

This report focuses on analyzing phishing emails, identifying indicators, classifying risks, and providing prevention guidelines.

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2. Objectives

The main objectives of this project are:

- To analyze real phishing email samples
- To identify common phishing indicators
- To classify emails as Safe, Suspicious, or Phishing
- To explain phishing attacks in simple, non-technical language
- To create awareness guidelines for users

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3. About Phishing Attacks

Phishing attacks aim to deceive users into performing harmful actions such as:

- Clicking malicious links
- Downloading infected attachments
- Sharing confidential data

Most phishing attacks succeed not because of weak systems, but because users are unaware of how to identify fake emails.

Common Types of Phishing:

- Email Phishing
- Spear Phishing (targeted attacks)
- Clone Phishing
- Whaling (targeting executives)

4. Methodology (Step-by-Step Process)

The following steps were followed in this project:

1. Collect phishing email samples from public datasets
2. Analyze email headers using tools
3. Inspect sender domain and URLs
4. Identify phishing indicators
5. Classify email risk
6. Document findings
7. Create awareness guidelines

5. Tools Used

- Public phishing datasets (Kaggle, SpamAssassin)
- Email Header Analyzer (Google Admin Toolbox, MXToolbox)
- Browser tools (Inspect Element, URL hover)
- Documentation tools (MS Word / PDF)

6. Phishing Email Analysis (Realistic Dataset-Based Examples)

Example 1: PayPal Spoofing Email

- Subject: Your PayPal Account Has Been Limited
- Sender: service@paypa1.com
- Analysis:
- Domain uses “1” instead of “l”
- Urgent message to verify account
- Contains fake login link
- Classification: Phishing

Example 2: Office 365 Credential Attack

- Subject: Password Expiry Notice
- Sender: admin@office365-security.com
- Analysis:
- Fake domain pretending to be Microsoft
- Urgency: “expires today”
- Redirects to phishing login page
- Classification: Phishing

Example 3: DHL Delivery Scam

- Subject: Delivery Failed
- Sender: support@dhl-delivery.net
- Analysis:
- Unexpected email
- Attachment contains malware
- Fake domain
- Classification: Phishing

Example 4: Bank Alert Scam

- Subject: Your Account Will Be Blocked
- Sender: security@bank-alerts.net
- Analysis:
- Threatening tone
- Requests login verification
- Suspicious link
- Classification: Suspicious / Phishing

Example 5: Lottery Scam

- Subject: You Won ₹10,00,000
- Sender: lottery@winnerpromo.xyz
- Analysis:
- Too good to be true
- Requests personal information
- Classification: Phishing

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7. Phishing Indicators

Common signs of phishing emails include:

- Spoofed or fake sender address
- Suspicious or shortened URLs
- Urgent or threatening language
- Unexpected attachments
- Poor grammar and spelling
- Requests for sensitive information

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8. Email Classification

Emails are categorized into:

1. Safe

- Verified sender
- No suspicious links
- Legitimate content

2. Suspicious

- Minor inconsistencies
- Unknown sender
- Needs verification

3. Phishing

- Fake domain
- Malicious links
- Attempts to steal data

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9. Prevention & Awareness Guidelines

Users should follow these practices:

- Always verify sender email address
- Hover over links before clicking
- Do not download unknown attachments
- Never share passwords or OTPs
- Enable two-factor authentication
- Report suspicious emails immediately

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10. Conclusion

Phishing remains a major cybersecurity threat due to human error and lack of awareness. This project demonstrated how phishing emails can be analyzed and detected using simple techniques.

By identifying key indicators and educating users, organizations can significantly reduce the risk of phishing attacks. Awareness and vigilance are the most effective defenses against such threats.

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11. References

- Kaggle Phishing Email Dataset
- SpamAssassin Public Dataset
- Google Admin Toolbox (Message Header Analyzer)
- MXToolbox Email Header Analyzer
